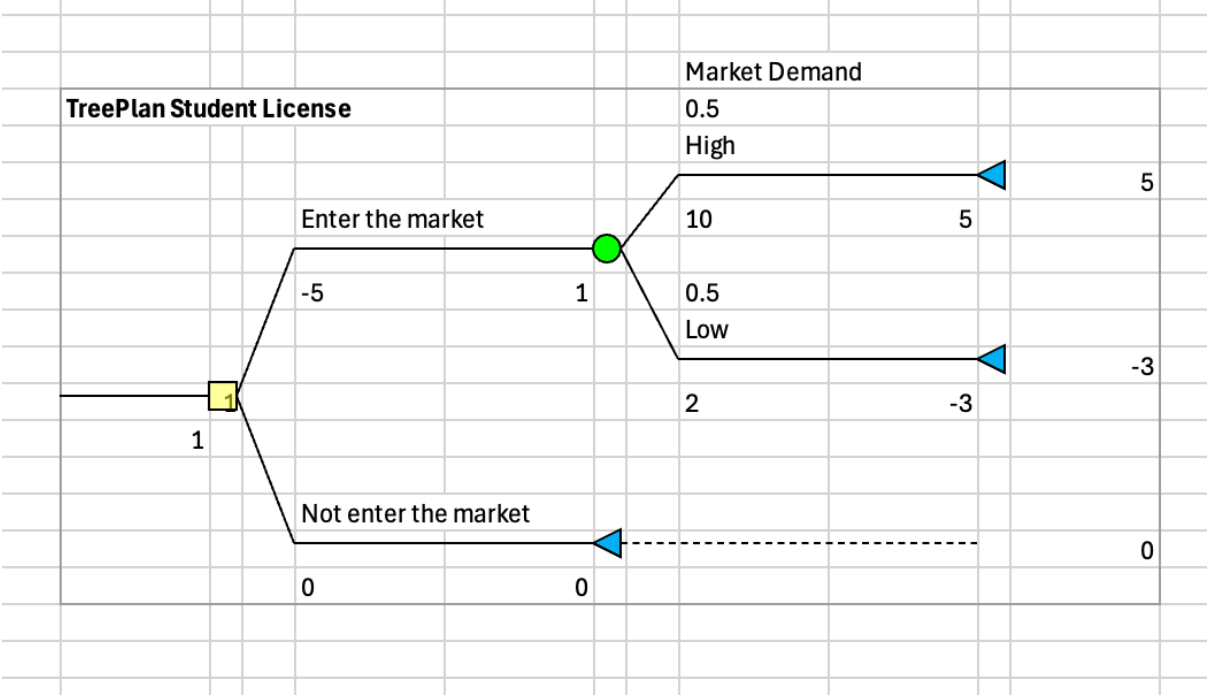
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HW 2

Part 1:

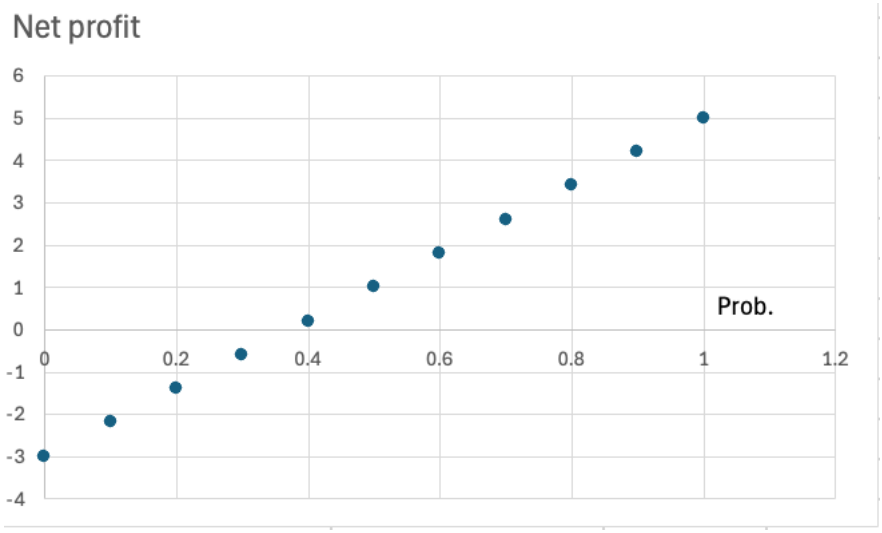
1. Optimal strategy is to enter the market since expected profit is 1 million dollars.



2. The range where optimal strategy changes if when the probability of high market demand falls between 0.3-0.4 which is the break-even point. If the probability of high market demand is over 0.4, the expected profit is positive, and Management can enter the market.

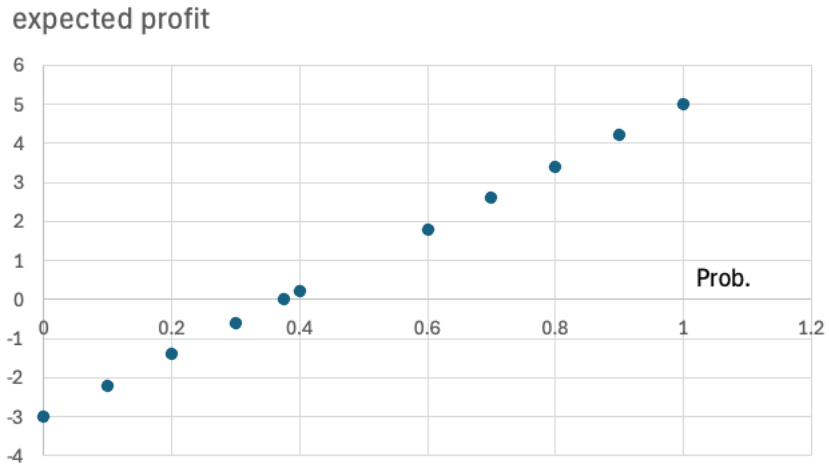
Prob(High Market demand)	Net profit
0.5	1
0.1	-2.2
0.2	-1.4
0.3	-0.6
0.4	0.2
0	-3
0.6	1.8
0.7	2.6
0.8	3.4
0.9	4.2
1	5

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3. The value of probability of high demand at which the decision changes from not entering the market is 0.375125 where expected profit is approximately 0.

Prob(High Market demand)	expected profit
0.375125	0.001
0.1	-2.2
0.2	-1.4
0.3	-0.6
0.4	0.2
0	-3
0.6	1.8
0.7	2.6
0.8	3.4
0.9	4.2
1	5



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Part 2:

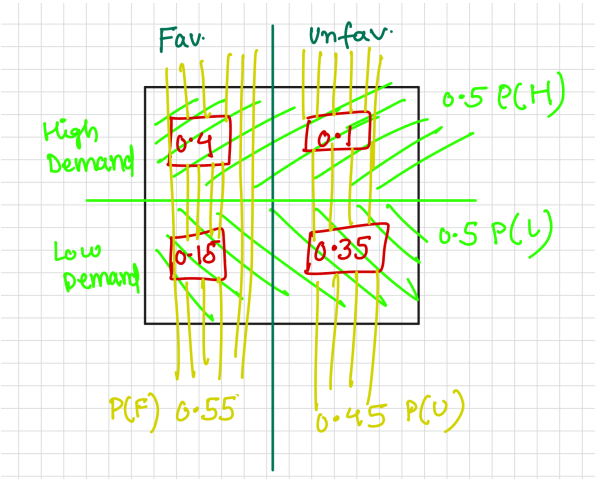
1. Given:
   P(H) = 0.5
   P(L) = 0.5
   P(F | H) = 0.8
   P(F | L) = 0.3

   P(A | B) = P(A, B) / P(B)

   So-> P(F,H) = 0.4
   P(F,L) = 0.15

1. P(F) = 0.4+0.15 = 0.55
2. P(U) = 1 – P(F) = 0.45
3. P(H | F) = 0.4/0.55 = 0.727
4. P(L | F) = 0.15/0.55 = 0.272
5. P(H | U)
   P(H,U) = P(H) – P(H,F) = 0.5 – 0.4 = 0.1
   P(H | U) = 0.1/0.45 = 0.2222

6. P(L | U)
   P(L,U) = P(L) – P(L,F) = 0.5 – 0.15 = 0.35
   P(L | U) = 0.35/0.45 = 0.77778



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2. Expected value without information: 1
   Expected value with information: 1.55
   VOI = EVSI = 1.55-1 = 0.55 < 1 million dollars required to invest in the study, the firm **should not** commission the study. We get 1 million dollars of expected profit without this study.

